

S VISHNU KOUSHIK

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Career Objective

Computer Science undergraduate with hands-on experience building **full-stack web applications** and **AI-driven solutions**, including **Retrieval-Augmented Generation (RAG)** pipelines and **NLP systems**. Proficient in **React**, **Node.js**, **Python**, **REST APIs**, and **Data Structures Algorithms**, with experience designing and deploying **production-ready applications**. Passionate about combining **full-stack engineering** with **Generative AI** to build scalable and intelligent software solutions.

Education

Vellore Institute of Technology – Amaravati *Sept 2023 – May 2027*
B.Tech in Computer Science Engineering

Tirumala Junior College *Sept 2021 – May 2023*
Board of Intermediate Education – MPC

Sri Chaitanya Techno School *May 2021*
Secondary School Certificate (SSC)

Projects

RoleFlow *MERN, Chrome Extension, MongoDB, REST API*

[Live Demo](#) — [GitHub](#)

- Developed a **full-stack job application** tracking platform that helps students and job seekers organize applications across multiple job portals from a centralized dashboard.
- Built a **Chrome Extension** to capture job details such as company, role, location, source, URL, and application date directly from job postings and sync them with the backend.
- Implemented secure **JWT authentication**, **application CRUD operations**, search/filtering, status tracking
- Designed **RESTful APIs with Node.js/Express and MongoDB/Mongoose**, with user-level authorization and structured application-event history.

CashCompass - CashManager *React.js, Node.js, Express.js, MongoDB, Chart.js*

[Live Demo](#) — [GitHub](#)

- Architected a scalable **RESTful API** using **Node.js/Express** and **MongoDB/Mongoose**, optimizing data retrieval efficiency by **30%** through schema indexing and lightweight payloads.
- Enhanced dashboard loading metrics by **15%** by developing responsive data visualizations via **Chart.js**.
- Deployed frontend on **Vercel** and backend on **Render**, managing CI/CD pipelines, environment variables, and CORS security policies.
- Implemented secure user authentication using **JWT** and encrypted password hashing, while designing category-wise transaction filtering and automated monthly budget threshold alerts.

RAG-Application - Ongoing *Python, FastAPI, LangChain, ChromaDB, React, OpenAI API*

[GitHub](#)

- Developing a **Retrieval-Augmented Generation (RAG)** system to answer questions from GATE Computer Science study material.
- Reduced semantic text-query search latency to under **200ms** by engineering an automated document ingestion pipeline utilizing LangChain, chunking text with **all-MiniLM-L6-v2**, and indexing dense vectors inside ChromaDB.

Technical Skills

Languages: Java, JavaScript, HTML, CSS **Databases:** MongoDB, Mongoose ODM

Frontend: React.js, Chart.js **Backend:** Node.js, Express.js, REST APIs, JWT

Tools: Git, Vercel, Render, Streamlit

Certifications

- Mastering Data Structures and Algorithms** – Faceprep *Arrays, Strings, Linked-Lists, Trees*
- Learning Full-Stack JavaScript Development** – LinkedIn *MongoDB, Node and React*
- Local AI: Build a RAG Model from Scratch with Open-Source Tools** – LinkedIn